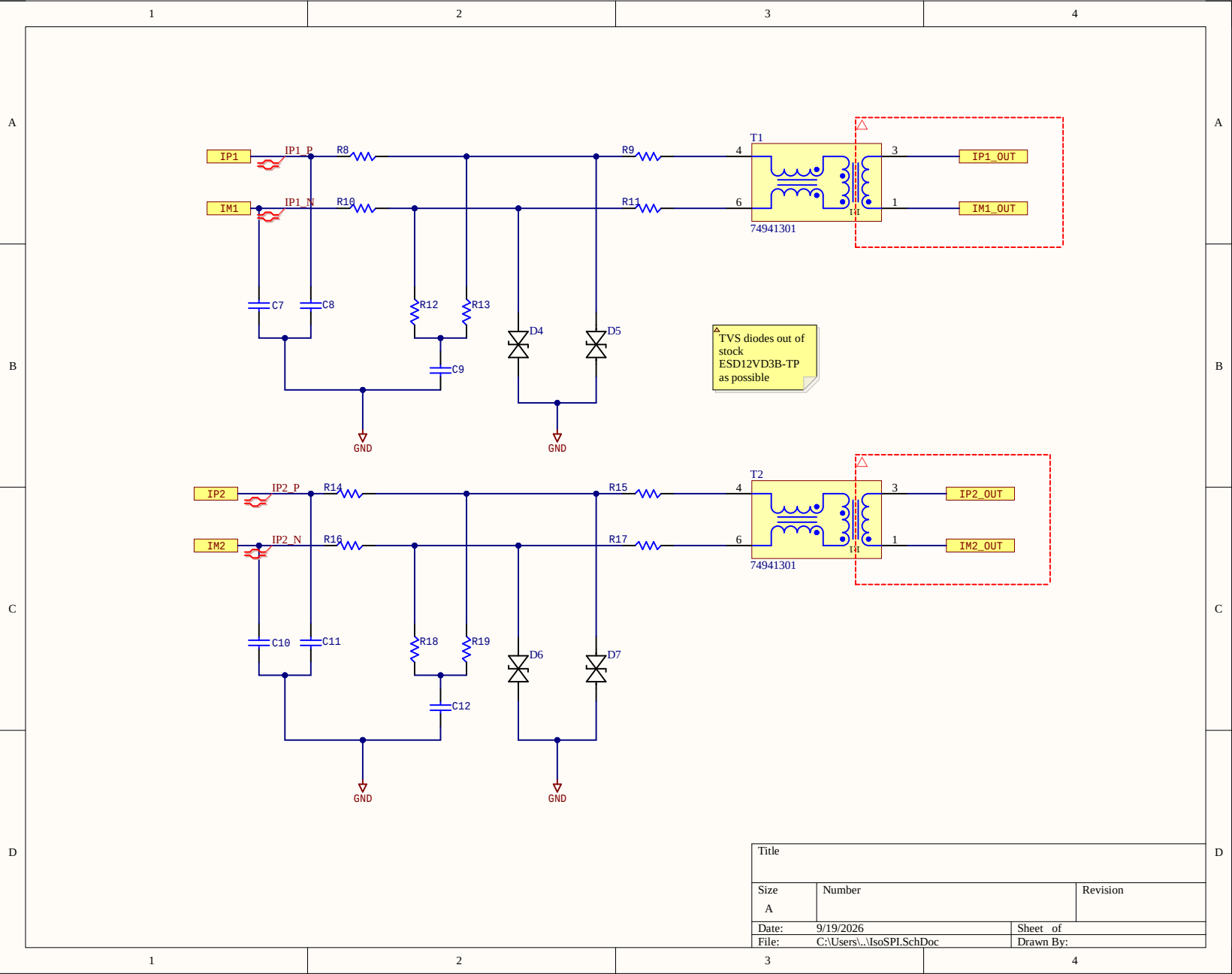
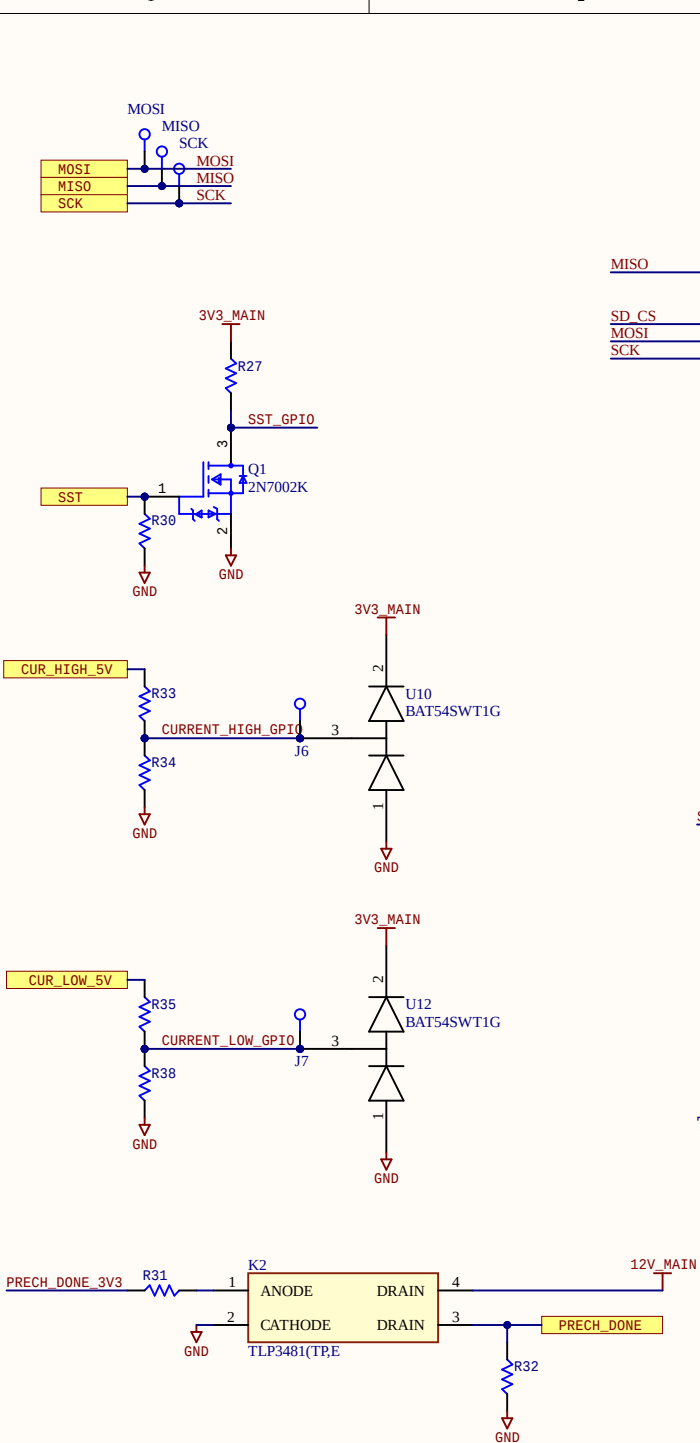
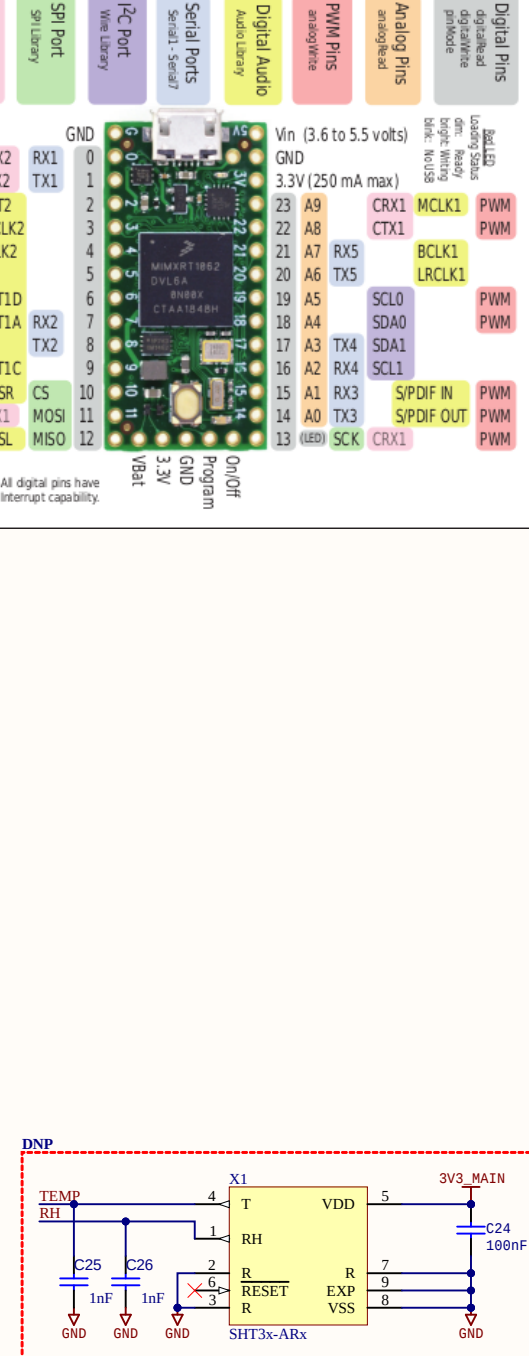
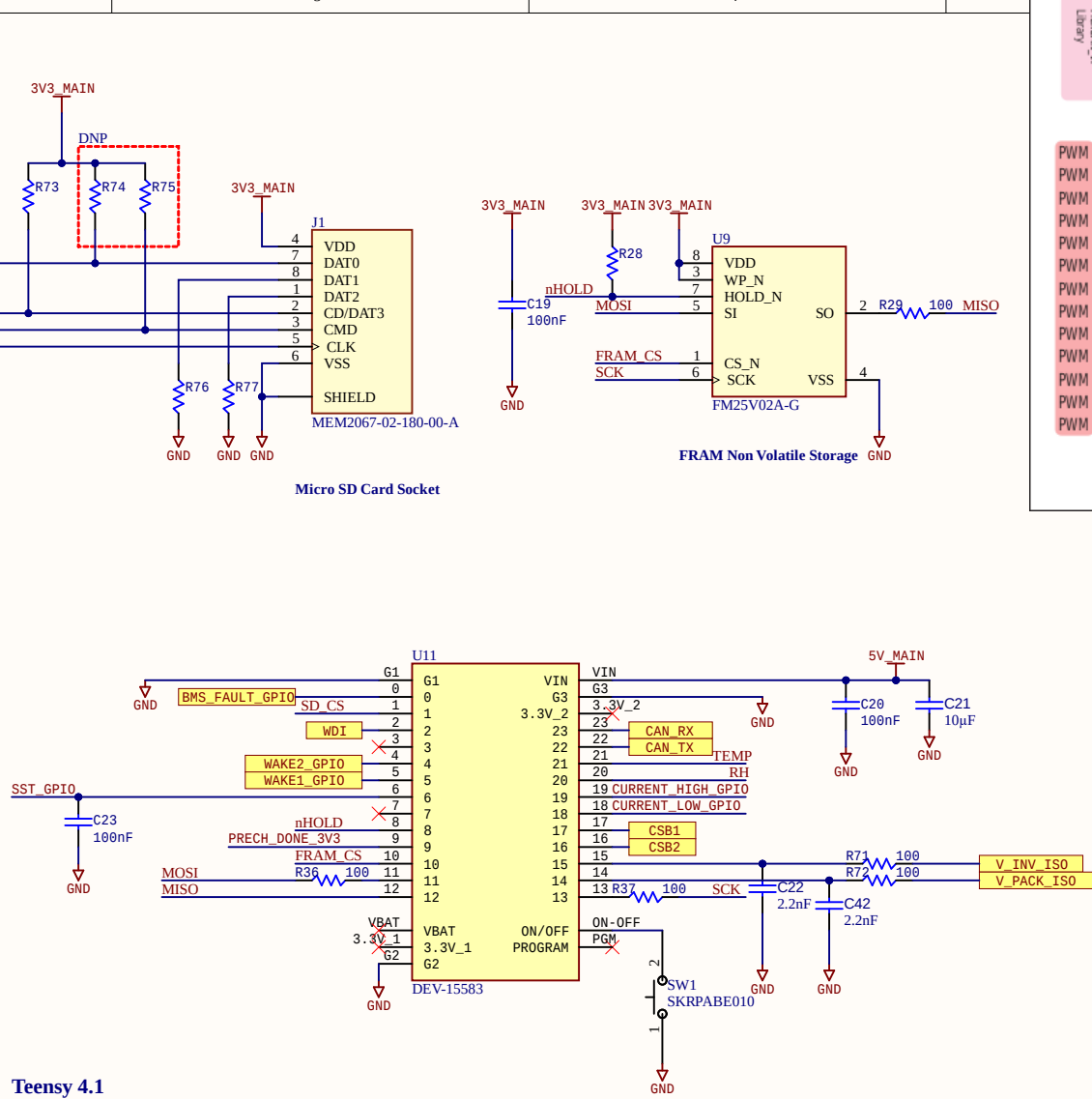
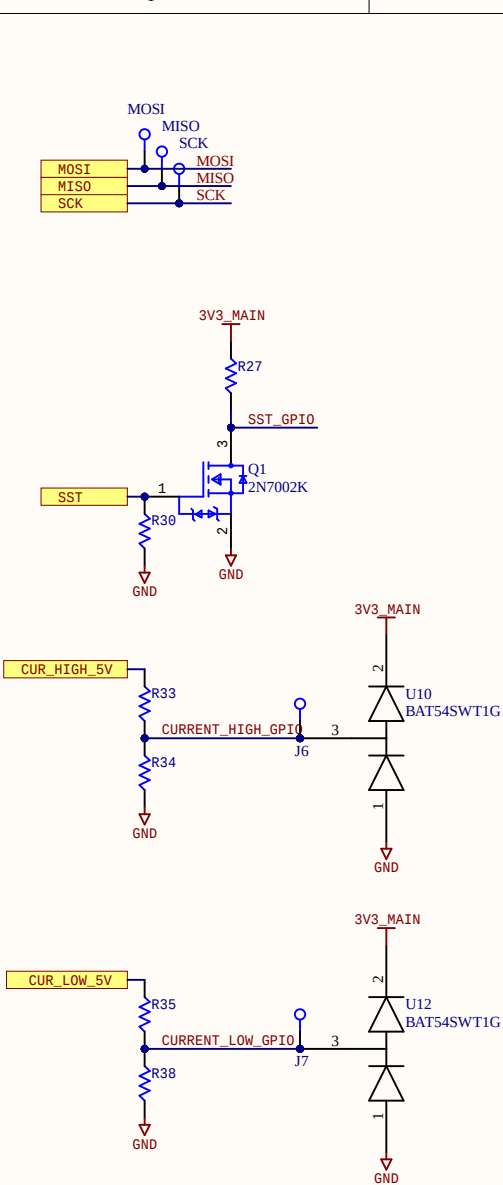
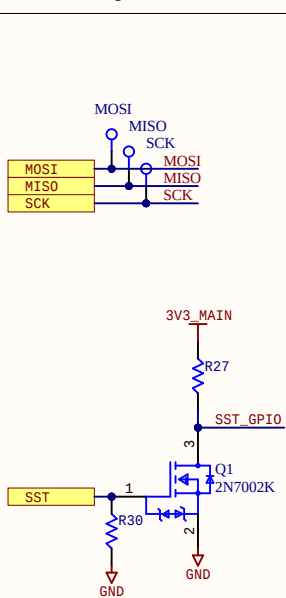
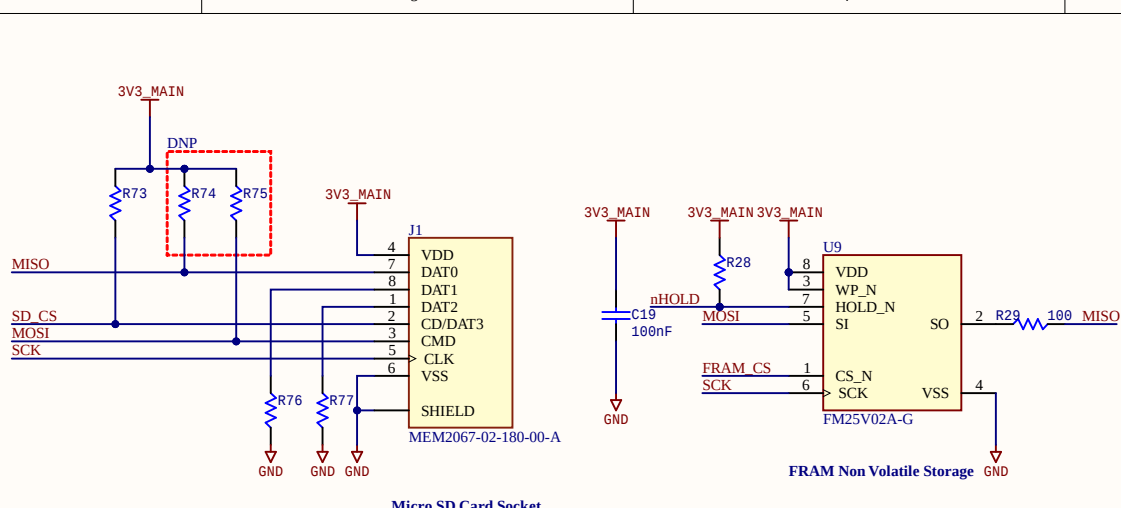
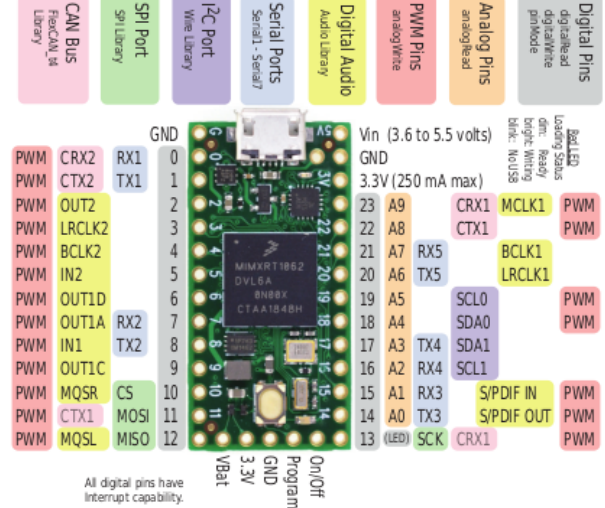




32 Bit Arduino-Compatible Microcontroller

www.pjrc.com/teensy



Interfacing Signals

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n Descriptions	
Description	
SPI Mode Select. These pins select SPI controller mode (MSTR connected to VDD5 and MSTR2 connected to VDD52) or peripheral mode (MSTR and MSTR2 connected to GND).	
SPI Controller Outputs (Controller Mode) or Peripheral Inputs (Peripheral Mode).	
SPI Controller Inputs (Controller Mode) or Peripheral Outputs (Peripheral Mode).	
SPI Clock Inputs (Controller) or Outputs (Peripheral).	
Active Low SPI Chip-Select Inputs (Controller Mode) or Outputs (Peripheral Mode).	
SPI Power-Supply Inputs (1.7 V to 5.5 V).	
LPCM Interrupt Current-Limited Outputs.	
Device Wake-Up State Current-Limited Outputs or Inputs.	
Ground. The GND pins must be shorted together external to the IC.	
Isolated Interface Minus Inputs/Outputs.	
Isolated Interface Plus Inputs/Outputs.	
isoSPI Power-Supply Inputs (3.0 V to 5.5 V).	
High Voltage Power-Supply Inputs (3.0 V to 30 V).	
Multilevel, Resistor Set SPI PHA/POL Selection Inputs.	
Multilevel, Resistor Set LPCM Timeout Selection Inputs.	
Multilevel, Resistor Set Transceiver Mode Selection Inputs.	
Exposed Pad. Must be connected to GND.	

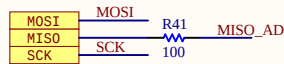
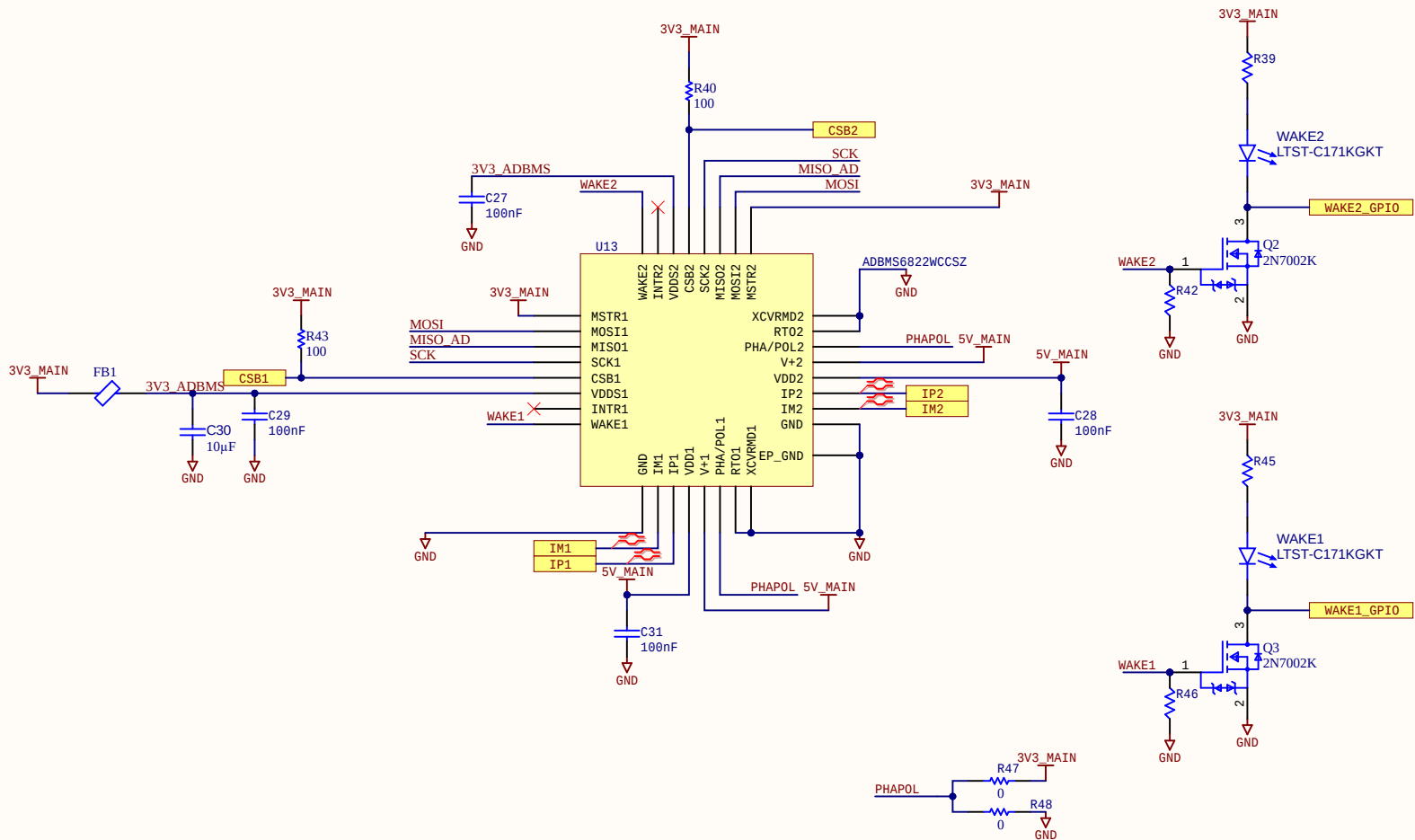


Table 21. XCVRMD Pin Transceiver Modes

Description	XCVRMD Pin Connection
Standard Bidirectional isoSPI	GND
Standard Bidirectional isoSPI with LPCM Timeout Monitor Support	20 kΩ resistor to GND
4 Mbps Unidirectional	100 kΩ resistor to GND
2 Mbps with 1-Bit Latency	VDD5

MSTR = 0 -> PERIPHERAL MODE
MSTR = 1 -> CONTROLLER MODE

MSTR -> Master (mode select)
PICO -> Peripheral In, Controller Out
POCI -> Peripheral Out, Controller In
SCK -> Serial Clock
CS -> Chip Select
VDD5 -> SPI power supply
INTR -> Interrupt
WAKE -> Wake pin
GND -> Ground
IM -> IsoSPI Minus
IP -> IsoSPI Plus
PHAPOL -> Phase/Polarity (SPI idle L/H, SPI latch rise/fall)
RTO -> Resistor set Timeout
XCVRMD -> Transceiver Mode
EP -> Exposed

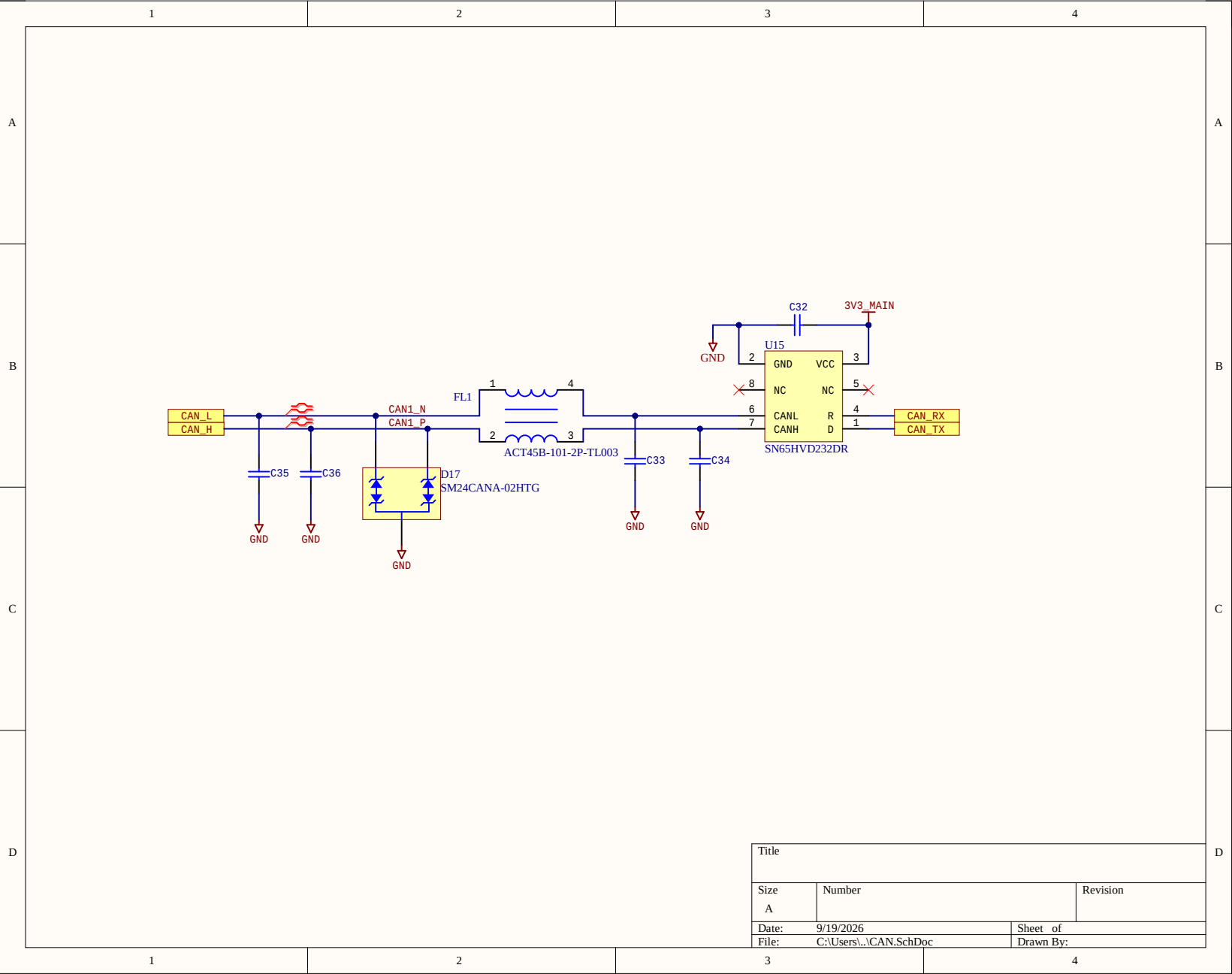


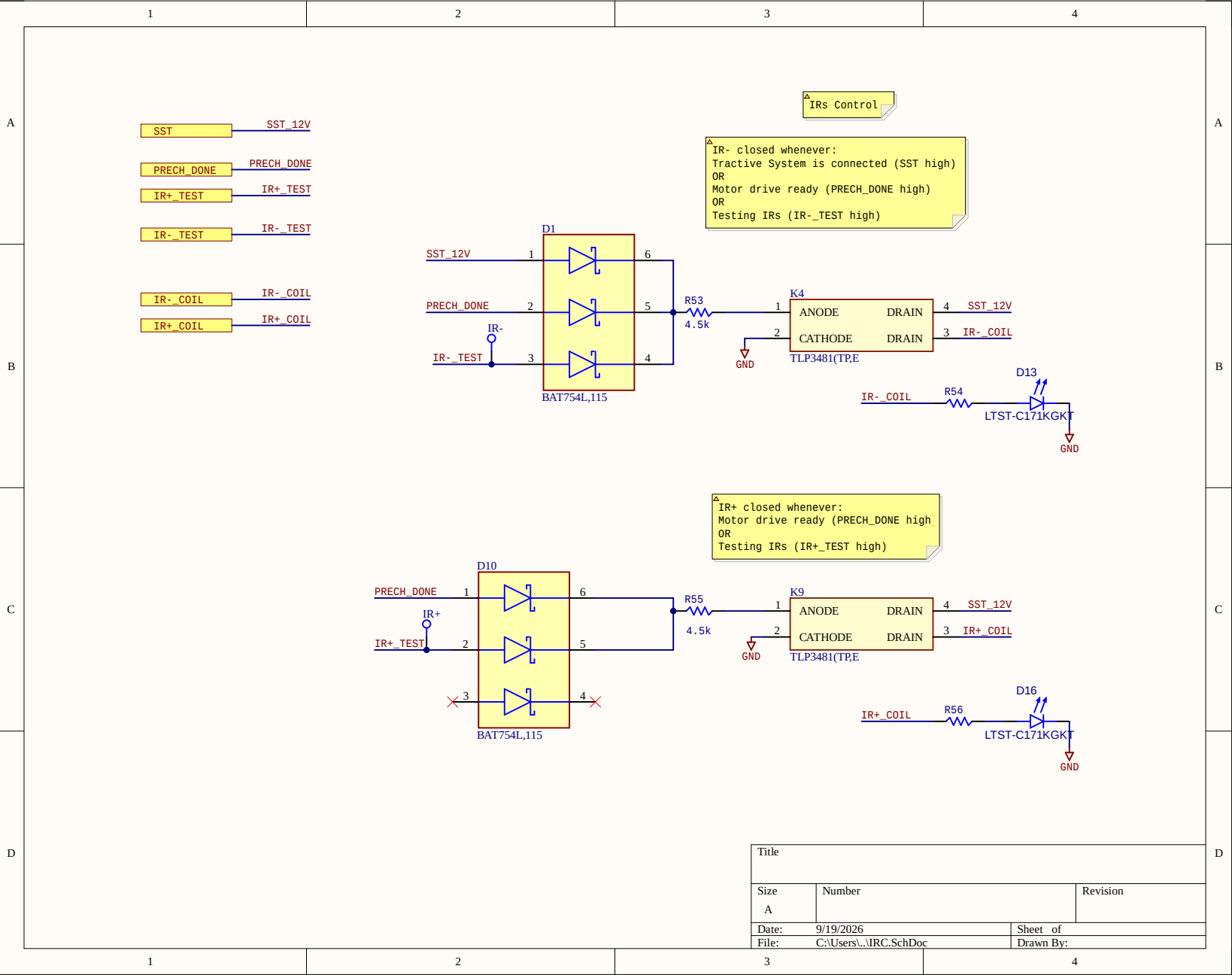
"ADBMS battery monitors use SPI Mode 3" Page 19

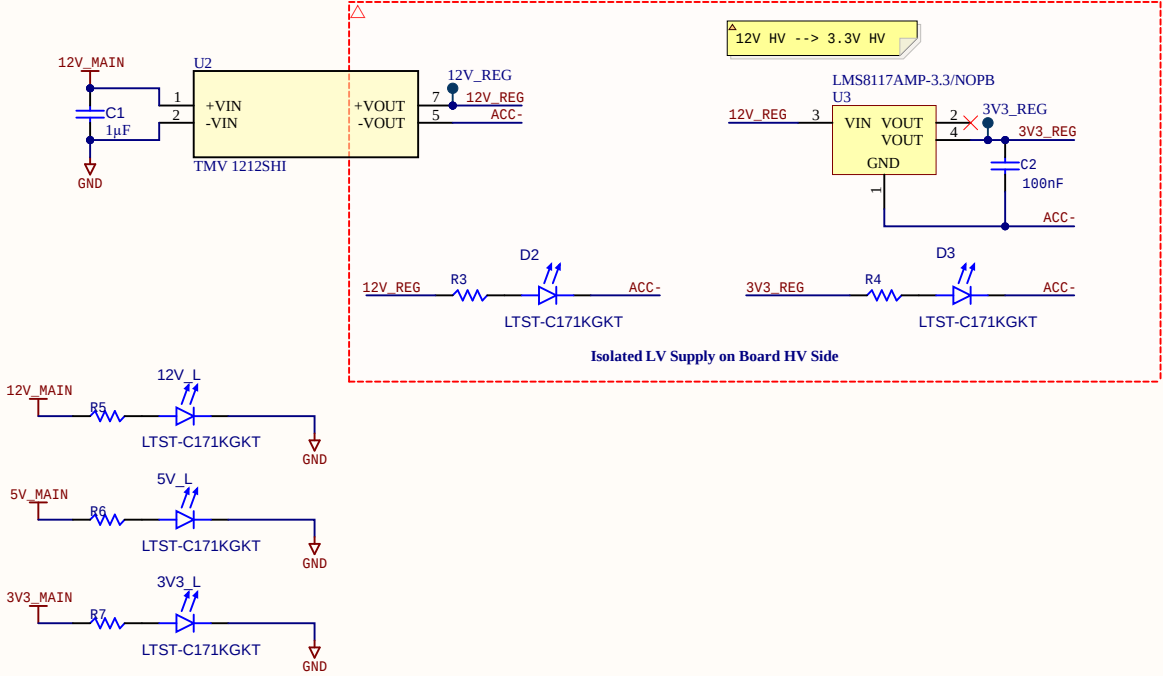
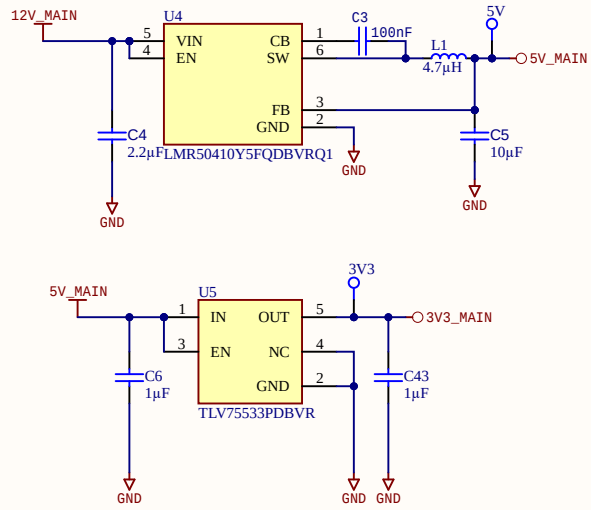
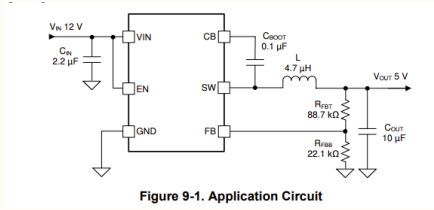
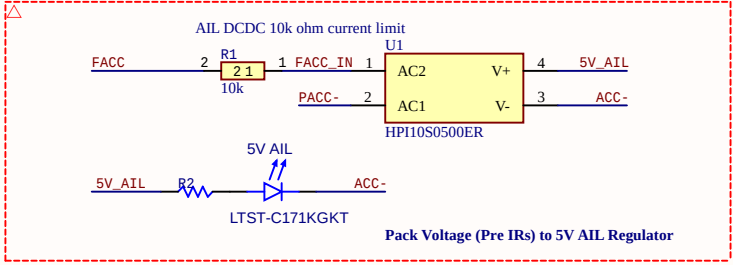
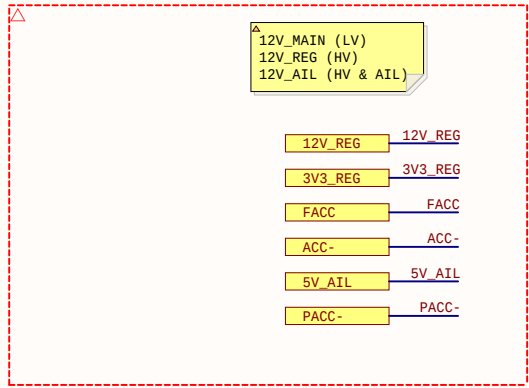
Table 20. PHAPOL SPI Modes

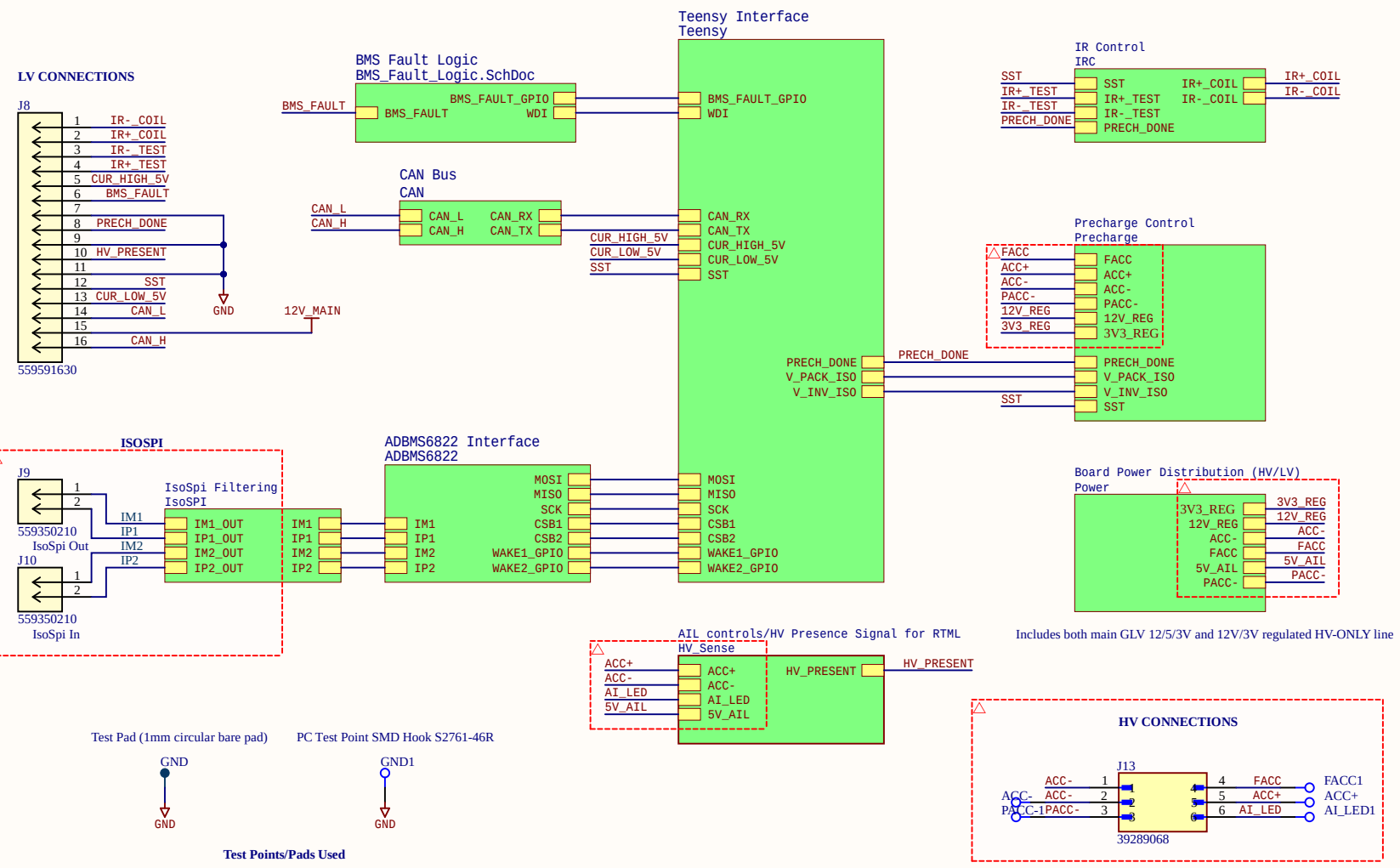
SPI Mode	POL	PHA	Description	PHAPOL Pin Connection
0	0	0	SCK idles low, latches on rising (first) edge	GND
1	0	1	SCK idles low, latches on falling (second) edge	20 kΩ resistor to GND
2	1	0	SCK idles high, latches on falling (first) edge	100 kΩ resistor to GND
3	1	1	SCK idles high, latches on rising (second) edge	VDD5

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